Lecture 6: DUALITY

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September 15, 2026

Outline

1. Dual function
2. Dual problem
3. Interpretations
4. Optimality conditions
5. Perturbation and sensitivity analysis

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The Lagrangian

- Consider a (general) optimization problem:

$$\min. f_0(x) \tag{1}$$
$$\text{s.t.} f_i(x) \leq 0, i = 1, \ldots, m \tag{2}$$
$$h_i(x) = 0, i = 1, \ldots, p. \tag{3}$$

  where $x \in \mathbb{R}^n$, domain $\mathcal{D} = \cap_{i=0}^{m} \mathbf{dom} f_i \cap \cap_{i=1}^{p} \mathbf{dom} h_i$ is non-empty, and p^* is the optimal value.
- The Lagrangian $L : \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}^p$ is given by

$$L(x, \lambda, \nu) = f_0(x) + \sum_{i=1}^{m} \lambda_i f_i(x) + \sum_{i=1}^{p} \nu_i h_i(x)$$

- λ_i and ν_i are *dual variables* or *Lagrange multipliers* associated with constraints $f_i(x) \leq 0$ and $h_i(x) = 0$, respectively. Vectors λ, ν are Lagrange multiplier associated with above optimization problem.
- Lagrangian function takes the constraints into account by augmenting the objective function with a weighted sum of the constraints.

Lagrange dual function

- *Lagrange dual function* (or just *dual function*) $g : \mathbb{R}^m \times \mathbb{R}^p \to \mathbb{R}$ is given by

$$g(\lambda, \nu) = \min_{x \in \mathcal{D}} L(x, \lambda, \nu) = \min_{x \in \mathcal{D}} \left(f_0(x) + \sum_{i=1}^{m} \lambda_i f_i(x) + \sum_{i=1}^{p} \nu_i h_i(x) \right)$$

- The dual function yields *lower bounds* on the optimal value p^* of the problem (3): For any $\lambda > 0$ and any ν we have $g(\lambda, \nu) \leq p^*$ (see p.216).

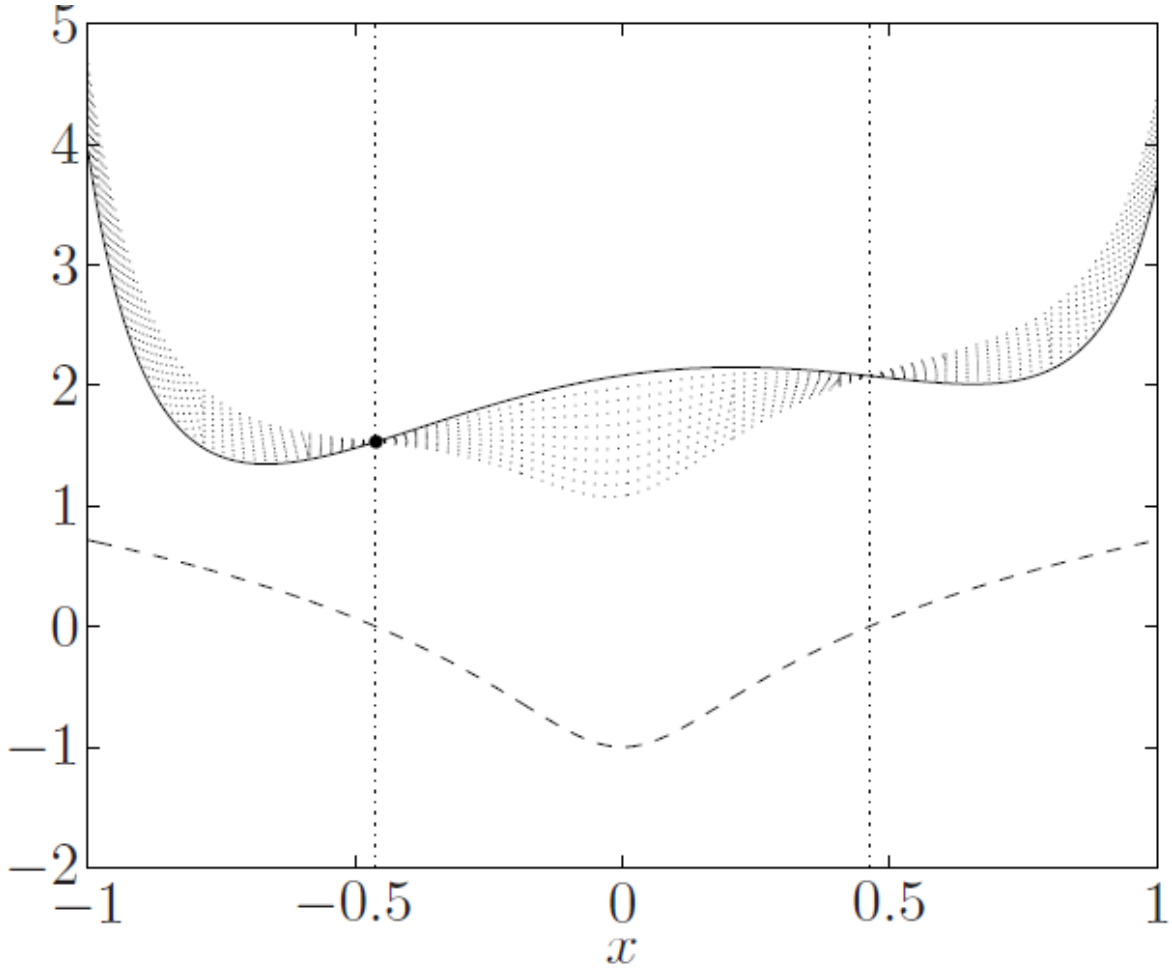


Figure 5.1 *Lower bound from a dual feasible point.* The solid curve shows the objective function f_0, and the dashed curve shows the constraint function f_1. The feasible set is the interval $[-0.46, 0.46]$, which is indicated by the two dotted vertical lines. The optimal point and value are $x^\star = -0.46$, $p^\star = 1.54$ (shown as a circle). The dotted curves show $L(x, \lambda)$ for $\lambda = 0.1,\ 0.2, \ldots, 1.0$. Each of these has a minimum value smaller than $p^\star$, since on the feasible set (and for $\lambda \geq 0$) we have $L(x, \lambda) \leq f_0(x)$.

Least-squares solution of linear equations

$$\begin{aligned} \text{min. } & x^T x \\ \text{s.t. } & Ax = B \end{aligned}$$

where $A \in \mathbb{R}^{p \times n}$.

- Lagrangian: $L(x, \nu) = x^T x + \nu^T (Ax - b)$, a convex quadratic function.
- Dual function: $g(\nu) = \min_x L(x, \nu)$. Solution of $\nabla_x L(x, \nu) = 2x + A^T \nu = 0$ is $x = -(1/2)A^T \nu$.
- Rewrite dual function: $g(\nu) = -(1/4)\nu^T A A^T \nu - b^T \nu$, a concave quadratic function (AA^T is positive definite). The lower bound property $g(\nu) \leq \min\{x^T x | Ax = B\}$

Linear Programming

$$\begin{aligned} \min\ & c^T x \\ \text{s.t. } & Ax = B \\ & x \geq 0. \end{aligned}$$

Lagrangian:

$$L(x, \lambda, \nu) = c^T x - \sum_{i=1}^{n} \lambda_i x_i + \nu^T (Ax - b) = -b^T \nu + (c + A^T \nu - \lambda)^T x.$$

Dual function:

$$\begin{aligned} g(\lambda, \nu) &= \min_x L(x, \lambda, \nu) = -b^T \nu + \min_x (c + A^T \nu - \lambda)^T x \\ &= \begin{cases} -b^T \nu \ \textit{if}\ \ c + A^T \nu - \lambda = 0 \\ -\infty, \textit{otherwise} \end{cases} \end{aligned}$$

The non-trivial lower bound is $-b^T \nu$ when $\lambda \geq 0$ and $c + A^T \nu - \lambda = 0$.

Conjugate function

- Let $f : \mathbb{R}^n \to \mathbb{R}$. The *conjugate* of the function f, $f^* : \mathbb{R}^n \to \mathbb{R}$ is defined as

$$f^*(y) = \max_{x \in \mathbf{dom} f} (y^T x - f(x)).$$

- f^* is a convex function since it is the pointwise supremum of a family of convex (indeed, affine) functions of y.

Conjugate function - examples

- $f(x) = ax + b$: the function $yx - ax - b$ is bounded if and only if y = a, hence, the domain of conjugate function is $\{a\}$, and $f^*(a) = -b$.
- $f(x) = -\log x$: the function $xy + \log x$ is unbounded above if $y \geq 0$ and reaches maximum at $x = -1/y$ otherwise.
  Hence $f^*(y) = -\log(-y) - 1$, for $y < 0$ and 0 if $y = 0$.

Find the conjugate functions of

- $f(x) = x \log x$
- $f(x) = 1/x$
- $f(x) = 1/2x^T Qx$

Conjugate function - examples

- Log-sum-exp function $f(x) = \log(\sum_{i=1}^{n} e^{x_i})$.

$$f^*(y) = \begin{cases} \sum_{i=1}^{n} y_i \log(y_i) & \text{if } y \geq 0 \text{ and } 1^T y = 1 \\ \infty & \text{otherwise} \end{cases}$$

- Norm $f(x) = \|x\|$

$$f^*(y) = \begin{cases} 0 & \text{if } \|y\|_* \leq 1 \\ \infty & \text{otherwise} \end{cases}$$

The conjugate of a norm is the indicator function of the dual norm unit ball.

Conjugate function and dual function

$$\begin{aligned} \min\ & f_0(x) \\ \text{s.t. } & Ax \leq b \\ & Cx = d. \end{aligned}$$

$$\begin{aligned} g(\lambda, \nu) &= \min_x (f_0(x) + \lambda^T (Ax - b) + \nu^T (Cx - d)) \\ &= -b^T \lambda - d^T \nu + \min_x (f_0(x) + (A^T \lambda + C^T \nu)^T x) \\ &= -b^T \lambda - d^T \nu - f_0^*(-A^T \lambda - C^T \nu). \end{aligned}$$

Conjugate function and dual function (cont.)

- Equality constrained norm minimization.

$$\begin{aligned} &\min \ \|x\| \\ &\text{s.t. } Ax = b. \end{aligned}$$

$$g(\nu) = -b^T\nu - f_0^*(-A^T\nu) = \begin{cases} -b^T\nu & \text{if } \|A^T\nu\|_* \le 1 \\ -\infty & \text{otherwise} \end{cases}.$$

- Entropy maximization

$$\begin{aligned} &\min \ f_0(x) = \sum_{i=1}^n x_i \log x_i \\ &\text{s.t. } Ax \le b \\ &\quad 1^T x = 1. \end{aligned}$$

$$g(\lambda,\nu) = -b^T\lambda - \nu - \sum_{i=1}^n e^{-a_i^T\lambda-\nu-1} = -b^T\lambda - \nu e^{-\nu-1}\sum_{i=1}^n e^{-a_i^T\lambda}.$$

Exercise

- Problems 5.1, 5.3, page 273.
- Prob. 5.5, 5.13, 5.14

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1. Dual function
2. Dual problem
3. Interpretations
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Dual problem

- For each pair (λ, ν) with $\lambda \geq 0$, the Lagrange dual function gives a lower bound on the optimal value p^* of the optimization problem (3).
- Consider the *dual problem*:

$$\begin{aligned} &\max\ g(\lambda, \nu) \\ &\text{s.t. } \lambda \geq 0. \end{aligned}$$

- (λ^*, ν^*) is dual optimal, or optimal Lagrange multipliers.

Linear programming

$$\begin{aligned} \min\ & c^T x \\ \text{s.t. } & Ax = B \\ & x \geq 0. \end{aligned}$$

- Dual function:

$$g(\lambda, \nu) = \min_x L(x, \lambda, \nu) = -b^T \nu + \min_x (c + A^T \nu - \lambda)^T x$$
$$= \begin{cases} -b^T \nu \ \textit{if}\ \ c + A^T \nu - \lambda = 0 \\ -\infty, \textit{otherwise} \end{cases}$$

- Dual function:

$$\begin{aligned} \max\ & -b^T \nu \\ \text{s.t. } & A^T \nu - \lambda + c = 0 \\ & \lambda \geq 0. \end{aligned}$$

or equivalently

Inequality form LP

$$\min\ c^T x$$
$$\text{s.t. } Ax \leq b$$

- Lagrangian: $L(x, \lambda) = c^T x + \lambda^T (Ax - b)$
- Dual function:

$$g(\lambda) = \min_x L(x, \lambda) = -b^T \lambda + \min_x (A^T \lambda + c)^T x$$
$$= \begin{cases} -b^T \lambda \ \textit{if}\ A^T + c\lambda = 0 \\ -\infty, \textit{otherwise} \end{cases}$$

- Dual problem:

$$\max -b^T \lambda$$
$$\text{s.t. } A^T \lambda + c = 0, \quad \lambda \geq 0.$$

Weak duality

- Let d^* be the optimal value of dual problem. The lower bound property yields $d^* \leq p^*$ for any original problem.
- $p^* - d^*$ is *optimal duality gap* of the primal problem (3), the gap between the optimal value and the best lower bound.

Strong duality and Slater's constraint qualification

- If the optimal duality gap is zero, *i.e.*, $d^* = p^*$, we say that *strong duality* holds.
- If the primal problem is convex, then the strong duality holds with mild condition (Slater's condition).
- Slater's condition: There exists a *strictly feasible* point such that $f_i(x) < 0, i = 1, \ldots, m$ and $Ax = b$.

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Saddle-point interpretation

- Consider the following problem

$$\begin{aligned} \min.\ & f_0(x) \\ \text{s.t.}\ & f_i(x) \leq 0, i = 1, \ldots, m. \end{aligned}$$

- Write the Lagrangian:
- The solution of primal problem $p^* = \min_x \max_{\lambda \geq 0} L(x, \lambda)$.
- The solution of dual problem $d^* = \max_{\lambda \geq 0} \min_x L(x, \lambda)$.
- Weak duality: $\max_{\lambda \geq 0} \min_x L(x, \lambda) \leq \min_x \max_{\lambda \geq 0} L(x, \lambda)$.

Economic interpretation

- Variable x denotes how the enterprise operates
- Objective $f_0(x)$ is cost of operating at x.
- Constraint $f_i(x) \leq 0$ limit on resources (labour, warehouse space), regulatory limit.
- Minimizing cost equivalents to maximizing profit. The optimal profit is $-p^*$.

Economic interpretation

- Consider scenarios in which the limits can be violated, by paying an additional cost which is linear in the amount of violation, measured by f_i.
- The payment made by the enterprise for the ith constraint is $\lambda_i f_i(x)$.
- The coefficient λ_i has the interpretation of the price for violating $f_i(x) \leq 0$;
- The total cost to the firm is $L(x, \lambda) = f_0(x) + \sum_{i=1}^{m} \lambda_i f_i(x)$. The firm will operate so as to minimize its total cost $L(x, \lambda)$, which yields a cost $g(\lambda)$.
- The optimal dual value, d^*, is the optimal cost to the enterprise under the *least favorable set of prices*.

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Complementary slackness

- Consider the optimization problem:

$$\begin{aligned} &\min. f_0(x) \\ &\text{s.t.} f_i(x) \leq 0, i = 1, \ldots, m \\ &\quad h_i(x) = 0, i = 1, \ldots, p. \end{aligned}$$

- Assume strong duality holds:

$$\begin{aligned} f_0(x^*) &= g(\lambda^*, \nu^*) \\ &= \min_x \left(f_0(x) + \sum_{i=1}^m \lambda_i^* f_i(x) + \sum_{i=1}^p \nu_i^* h_i(x) \right) \\ &\leq f_0(x^*) + \sum_{i=1}^m \lambda_i^* f_i(x^*) + \sum_{i=1}^p \nu_i^* h_i(x^*) \leq f_0(x^*). \end{aligned}$$

- $\sum_{i=1}^m \lambda_i^* f_i(x^*) = 0$, hence, $\lambda_i^* f_i(x^*) = 0$ for $i = 1, \ldots, m$.
- If $\lambda_i^* > 0$, then $f_i(x^*) = 0$. If $f_i(x^*) < 0$, then $\lambda_i^* = 0$

KKT optimality conditions for non-convex problems

Assume that $f_0, \ldots, f_m, h_1, \ldots, h_p$ are differentiable. Let x^* and (λ^*, ν^*) be any primal and dual optimal point with **zero duality gap**. They satisfy the *Karush-Kuhn-Tucker (KKT) conditions*:

$$
\begin{aligned}
f_i(x^*) &\leq 0, i = 1, \ldots, m \\
h_i(x^*) &= 0, i = 1, \ldots, p \\
\lambda_i^* &\geq 0, i = 1, \ldots, m \\
\lambda_i^* f_i(x^*) &= 0, i = 1, \ldots, m \\
\nabla f_0(x^*) + \sum_{i=1}^{m} \lambda_i^* \nabla f_i(x^*) + \sum_{i=1}^{p} \nu_i^* \nabla h_i(x^*) &= 0.
\end{aligned}
$$

KKT optimality conditions for convex problems

When the primal problem is convex, the KKT conditions are also sufficient for the points to be primal and dual optimal.
If any point $(\tilde{x}, \tilde{\lambda}, \tilde{\nu})$ satisfy the KKT conditions:

$$
\begin{aligned}
f_i(\tilde{x}) &\leq 0, i = 1, \ldots, m \\
h_i(\tilde{x}) &= 0, i = 1, \ldots, p \\
\tilde{\lambda}_i &\geq 0, i = 1, \ldots, m \\
\tilde{\lambda}_i f_i(\tilde{x}) &= 0, i = 1, \ldots, m \\
\nabla f_0(\tilde{x}) + \sum_{i=1}^{m} \tilde{\lambda}_i \nabla f_i(\tilde{x}) + \sum_{i=1}^{p} \tilde{\nu}_i \nabla h_i(\tilde{x}) &= 0.
\end{aligned}
$$

then $\tilde{x}$ and $(\tilde{\lambda}, \tilde{\nu})$ are primal and dual optimal with zero duality gap.

Examples

Equality constrained convex quadratic minimization.

$$\begin{aligned} &\min\ (1/2)x^T Px + q^T x + r \\ &\text{s.t. } Ax = b, \end{aligned}$$

where $P \in S_+^n$ (the convex problem). The KKT conditions for this problem are

$$Ax^* = b, Px^* + q + A^T \nu^* = 0.$$

$$\begin{bmatrix} P & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} x^* \\ \nu^* \end{bmatrix} = \begin{bmatrix} -q \\ b \end{bmatrix}.$$

Examples

Examples 5.3, 5.4, page 262.

Examples: Water-filling problem

Water-filling problem maximizing channels' capacity. x_i is the transmit power allocated to channel i and $\log(\alpha_i + x_i)$ is the channel capacity.

$$\begin{aligned} \min \ & -\sum_{i=1}^{n} \log(\alpha_i + x_i) \\ \text{s.t.} \ & \sum_{i=1}^{n} x_i = 1, \\ & x \geq 0, \end{aligned}$$

where $\alpha_i > 0$.
Lagrangian function

$$L(x, \lambda, \nu) = -\sum_{i=1}^{n} \log(\alpha_i + x_i) - \sum_{i=1}^{n} \lambda_i x_i + \nu(\sum_{i=1}^{n} x_i - 1)$$

KKT conditions:

$$x_i^* \geq 0, \sum_{i=1}^{n} x_i^* = 1,$$
$$\lambda_i^* \geq 0,$$
$$\lambda_i^* x_i^* = 0,$$
$$-1/(\alpha_i + x_i^*) - \lambda_i^* + \nu^* = 0, i = 1, \ldots, n.$$

Eliminate slack variables λ_i:

$$x^* \geq 0, \sum_{i=1}^{n} x_i^* = 1,$$
$$x_i^*(-1/(\alpha_i + x_i^*) + \nu^*) = 0,$$
$$-1/(\alpha_i + x_i^*) + \nu^* \geq 0, i = 1, \ldots, n.$$

Water-filling problem

If $\nu^\star \geq 1/\alpha_i$ (*i.e.*, $\alpha_i \geq 1/\nu^\star$), we must have $x_i^* = 0$. Otherwise, $\nu^\star > 1/(\alpha_i + x_i^\star)$, violate complementary slackness.
If $\nu^* < 1/\alpha_i$, we must have $x_i^* > 0$, hence, $x_i^* = 1/\nu^* - \alpha_i$.

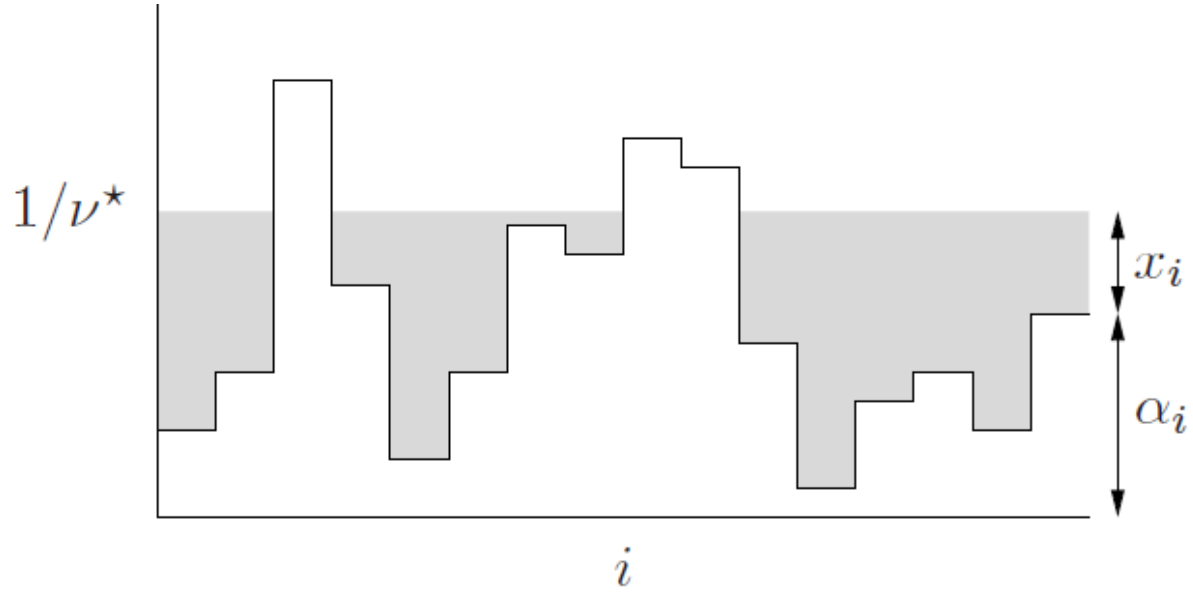


Figure 5.7 Illustration of water-filling algorithm. The height of each patch is given by α_i. The region is flooded to a level $1/\nu^\star$ which uses a total quantity of water equal to one. The height of the water (shown shaded) above each patch is the optimal value of $x_i^\star$.

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Perturbation and sensitivity analysis

- Perturbed problem

$$\begin{aligned} \min. & f_0(x) \\ \text{s.t.} & f_i(x) \leq u_i, i = 1, \ldots, m \\ & h_i(x) = v_i, i = 1, \ldots, p. \end{aligned}$$

- Define $p^*(u, v)$ as the optimal value of the perturbed problem.
- Let (λ^*, ν^*) be optimal for the dual of the unperturbed problem.
- A global inequality

$$p^*(u, v) \geq p^*(0, 0) - \lambda^{*T} u - \nu^{*T} w$$

Sensitivity interpretations

$$p^*(u, w) \geq p^*(0, 0) - \lambda^{*T} u - \nu^{*T} w.$$

- If λ_i^* is large and we tighten the ith constraint (i.e., choose $u_i < 0$), then the optimal value $p^*(u, w)$ is guaranteed to increase greatly.
- If ν_i^* is large and positive and we take $w_i < 0$, or if ν_i^* is large and negative and we take $w_i > 0$, then the optimal value $p^*(u, w)$ is guaranteed to increase greatly.
- If λ_i^* is small, and we loosen the ith constraint ($u_i > 0$), then the optimal value $p^*(u, w)$ will not decrease too much.
- If ν_i^* is small and positive, and $w_i > 0$, or if ν_i^* is small and negative and $w_i < 0$, then the optimal value $p^*(u, w)$ will not decrease too much.

Local sensitivity analysis

- Suppose that $p^*(u, w)$ is differentiable at $u = 0, w = 0$ and strong duality holds.
- We have the relation:

$$\lambda_i^* = -\frac{\partial p^*(0,0)}{\partial u_i}, \quad \nu_i^* = -\frac{\partial p^*(0,0)}{\partial w_i}$$

- The optimal Lagrange multipliers are exactly the local sensitivities of the optimal value with respect to constraint perturbations.
- $-\lambda^*$ is the slope of $p^*(u, w)$ near $u = 0$, $-\nu^*$ is the slope of $p^*(u, w)$ near $w = 0$

Summary

- Lagrangian, dual function, dual problem
- Optimality conditions (KKT)
- Perturbation and sensitivity analysis

References

- Convex optimization book by Stephen Boyd